Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 322 : Computer Peripheral & Interfacing Sessional

Project Proposal

Project Title : Smart Home
Submission Date : 21 January 2026

Submitted by,

1. **2102002**- Rusni Akter
2. **2102014** – Sadia Homayra Nowshin
3. **2102018** – Jannatul Mim
4. **2102047** – Puspita Baidya
5. **2102055** – Mst.Meherin Jahan Jui

Submitted To,

Sarna Majumder

Associate Professor,

Depart. of Computer and Communication Engineering, Patuakhali Science and Technology University.

Prof. Dr. Md Samsuzzaman

Professor,

Depart. of Computer and Communication Engineering,

Contents

1. Introduction ……3
2. Objectives ……3
3. Problem Statement ……3
4. Scope ……3
5. Methodology ……4
    5.1. System Architecture ……4
    5.2. Technology Stack ……4
6. Hardware Components ……5
7. Budget Estimation ……6
8. Work Plan (Timeline) ……8
9. Visual Models ……9
    9.1. Block Diagram ……9
    9.2. Flow Chart ……10
10. Future Plans ……11
11. Conclusion ……11

Smart Home

1. Introduction

Smart Home is an IoT-based home automation system that centralizes control of lighting, gate of garage and access through mobile and voice interfaces. Beyond standard automation, the system integrates advanced safety monitoring (fire/smoke), a **comprehensive Smart Aquarium management system**, a smart rooftop and an ML-powered **Smart Baby-Care** feature. By leveraging a dual-controller architecture (ESP32 and Raspberry Pi), the system balances real-time hardware response with complex machine learning inference. The system incorporates a Raspberry Pi-based live video streaming module that allows users to monitor their home environment in real-time via a mobile application

2. Objectives

- **Automate Household Tasks:** Control lighting, fans, and gates with secure manual overrides and gate of garage on/off by mobile.
- **Enhance Safety:** Implement real-time fire and gas detection with automated ventilation.
- **Smart Aquarium Management:** Monitor and maintain water quality (pH, TDS, temperature) and automate feeding/water level schedules.
- **ML-Based Baby Care:** Detect baby cries using acoustic AI to trigger soothing responses and notify parents.
- **Unified Monitoring:** Provide a single mobile interface for status logs, camera feeds, visual surveillance and automation rules.
- **Smart Rooftop:** provide a automatic and mobile controlled rooftop.

3. Problem Statement

Manual household management is inefficient and lacks integration. Critical safety incidents like gas leaks or fires require immediate, automated responses that manual systems cannot provide. Furthermore, caregivers face challenges in multi-tasking, often missing a baby's distress or neglecting precise environmental needs for aquatic life. A unified IoT system addresses these gaps by automating routine tasks, providing 24/7 safety surveillance, and offering specialized care for infants and pets. Traditional CCTV systems often operate in isolation, making it difficult for homeowners to get immediate visual confirmation during automated alerts like fire detection or baby distress.

4. Scope

- **Automation:** LDR-based garden lights, room utilities and **Smart Aquarium scheduling (feeding/lighting)**.
- **Access Control:** Mobile-controlled garage gates and multi-factor main gate access (RFID, PIN, Face Recognition).
- **Safety:** Fire/Smoke detection with active buzzer alarms and exhaust fan activation.
- **Baby Care:** Audio-based cry detection via ML, automated music playback, and push notifications.
- **Aquarium Monitoring:** Real-time tracking of **pH levels, Total Dissolved Solids (TDS), and water temperature**, plus water level depth detection.
- **Visual monitoring:** Implementation of live video streaming using a Raspberry Pi camera, including remote access via the mobile app and event-based visual monitoring**.**
- **Smart Rooftop:** provide a automatic and mobile controlled rooftop.

5. Methodology

The project follows an iterative cycle, integrating hardware modules into a unified rule engine. The Smart Aquarium uses calibrated pH/TDS sensors with an RTC for precise scheduling. The Raspberry Pi 4 serves as the intelligence hub, running a TensorFlow Lite model for baby cry detection while simultaneously hosting a Flask/OpenCV server to stream live CCTV footage to the mobile app via HTTP/RTSP.

5.1 System Architecture

- **Edge Controllers: ESP32** handles low-level GPIO (relays, simple sensors); **Raspberry Pi 4** manages ML inference, the camera feed, and high-level data processing.
- **Sensor Layer:**
    - *Home:* LDR, Flame, Smoke/Gas, DS18B20 (Temp),Rain
    - *Aquarium:* **pH Electrode, TDS Meter, Waterproof DS18B20, Water Level Sensor.**
    - *Input:* INMP441 Microphone, RFID, 4x4 Keypad.

- **Actuation Layer:** Relays (Lights/Fans), L298N Motor Drivers (Gates), Speakers (Baby-care), and DC-DC Buck Converters for stable power distribution.
- **Application Layer:** A mobile dashboard for real-time monitoring, event logging, and rule configuration.

5.2 Technology Stack

- **Hardware:** Raspberry Pi 4 (4GB), ESP-32S NodeMCU, PCF8563 RTC.
- **Firmware/Software:** Arduino (C++) for ESP32; Python for Raspberry Pi services.
- **Communication:** MQTT for low-latency messaging; HTTP for data logging.
- **Machine Learning:** MFCC feature extraction with a CNN/LSTM model deployed via TensorFlow Lite.
- **Power Management:** 12V to 5V/3V DC-DC Buck Step-down Converters to power various modules from a single source.

6.Hardware Components Breakdown

Category	Component	Purpose
Control	Raspberry Pi 4 & ESP32	Dual-core brain for ML and real-time I/O.
Aquarium	pH Electrode & TDS Meter	Chemical water quality monitoring.
Aquarium	Waterproof DS18B20	Submersible temperature tracking.
Aquarium	Water Level Sensor	Monitoring tank depth/overflow.
Scheduling	PCF8563 RTC Module	Keeping time for aquarium feeding/light cycles.
Baby Care	INMP441 Mic & CA6928 Amp	Audio capture and soothing sound playback.

Category	Component	Purpose
Safety	Flame & Smoke Sensors	Emergency detection.
Drive	L298N & 12V Gear Motor	Mechanical gate and utility control.
CCTV Camera	Raspberry Pi Camera / USB Webcam	Real-time video streaming and remote household monitoring.
Smart Rooftop	Rain sensor	

7.Budget Estimation

The following table summarizes the estimated hardware budget based on the prepared component list.

Item Description	Unit Price (BDT)	Qty	Total (BDT)
Raspberry pi 4(4GB)	14100	1	14100
Raspberry Pi Camera 5MP	740	1	740
MicroSD Card(32 GB, Class 10)	1000	1	1000
Power supply 5/3V	1200-1500	1	1200-1500
ESP32 ESP-32S 30P NodeMCU Development Board Wireless WiFi	427	1	427
DS18B20 Digital Thermometer	95	1	95
Ohm (Ω) 1/4w Resistors - Pack of 5 - Value: 100Ω	5	2	10
PCF8563 Real Time Clock RTC Module	190	1	190
L298N H-Bridge Dual Motor Driver, Stepper Motor Driver	168	1	168
Water Level Sensor Depth of Detection Water Sensor for Arduino	49	1	49

12V Fixed Output DC-DC Buck Step-down Converter	195	1	195
E-201/E-201C Rechargeable Composite pH Electrode	1500	1	1500
Waterproof DS18B20 Digital Thermal Probe or Sensor	209	1	209
TDS Meter V1.0 Module Water Quality Sensor for Arduino	1830	1	1830
PIR / Ultrasonic Sensor	90	1	90
Crystal Oscillator (Frequency)	20	1	20
Keypad Matrix Keypad (4x4 or 3x4)	70	1	70
RFID	160	1	160
Breadboard	120	1	120
Display	330	1	330
servo	120	4	360
wires	200		200
Flame Sensor Fire Detection Module	50	1	50
Passive buzzer	15	2	30
Rain sensor	80	1	80
Smoke $ gas detector sensro	160	1	160
5V Cooling Fan - 40 x 40 x 7mm	190	1	190
Photosensitive LDR Sensor Module	75	1	75
LED light	5	10	50
2 Channel 5V Relay Board Module	148	2	296
YL-69 Soil Hygrometer Humidity & Soil Moisture Detection Sensor	70	1	70
DC water pump motor	150	1	150
GY-9960LLC APDS-9960(gesture sensor)	550	1	550
12V 60 RPM DC Gear Motor(g)	150	1	150
L298N H-Bridge Dual Motor Driver, Stepper Motor Driver(g)	185	1	185
INMP441 Omnidirectional Microphone Module Original (for sound input)(b)	590	1	590
Stereo Enclosed Speaker 3W 8Ω(b)	390	1	390

Digital Power Channel Amplifier Board 5V Bluetooth Speaker Audio CA6928(b)	80	1	80
Grand Total (Estimated)			26,159 – 26,459

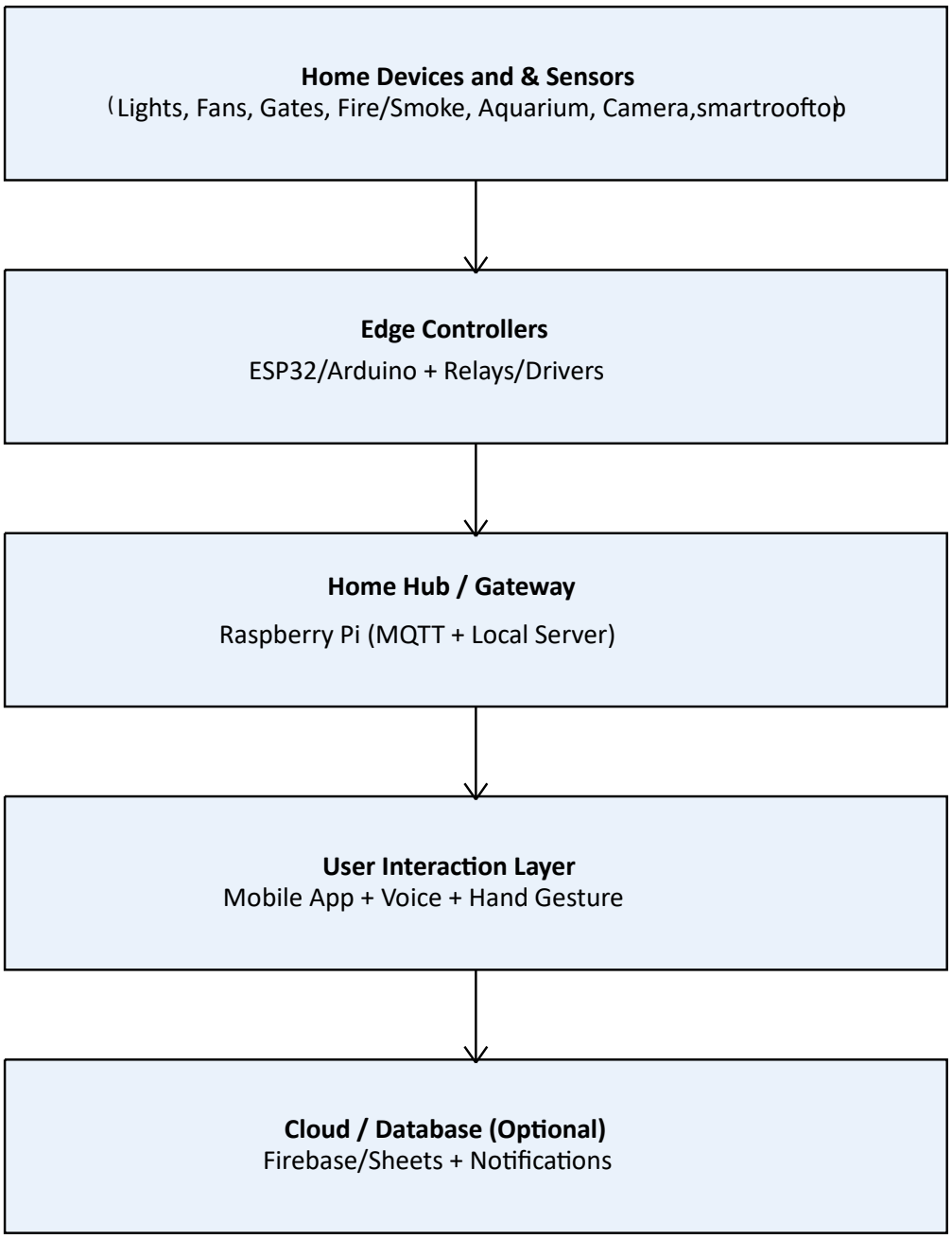
8.Work Plan (Timeline)

The work plan is organized into design, implementation, integration, and validation phases. Parallel development is used where possible (e.g., app UI and hardware assembly).

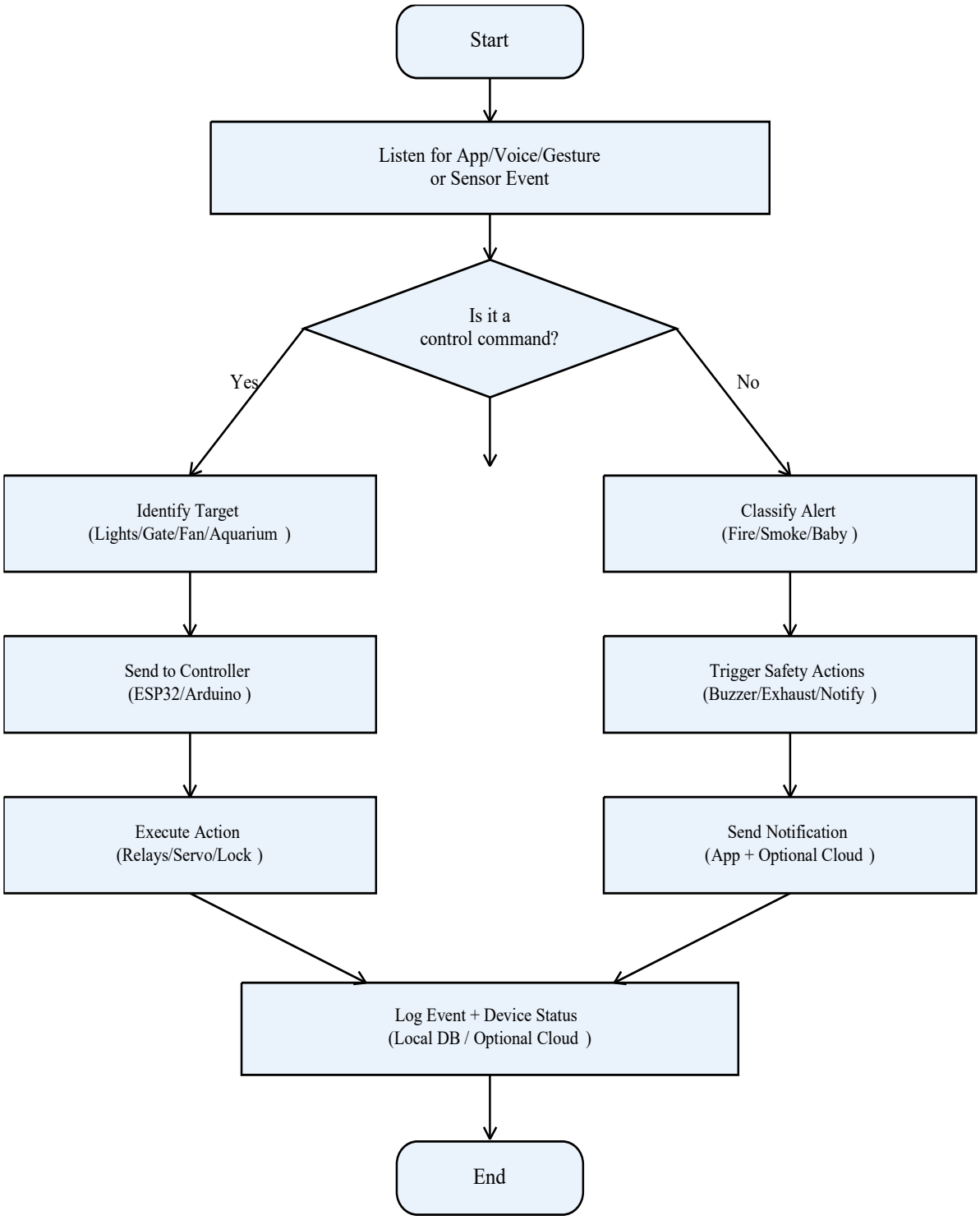
Phase / Task	Month 1	Month 2	Month 3	Month 4
Requirements & detailed design	✓			
Procurement & bench setup	✓			
Core automation (lights/fans/schedules)	✓	✓		
Gate/garage control + access control		✓	✓	
Safety module (fire/smoke response)		✓	✓	
Smart aquarium automation			✓	
ML baby cry detection + music trigger			✓	✓
Voice/gesture control (prototype)			✓	✓
Integration testing & final documentation				✓

9.Visual Models

9.1.Block Diagram



9.2. Flow Chart



10.Future Plans

- **Encrypted Access & Security:** Implement industry-standard encryption (e.g., AES-256) for the mobile app and establish role-based access control (RBAC) to distinguish between "Admin" (parents) and "User" (family/children) permissions.
- **Advanced Computer Vision:** Upgrade face recognition with high-definition camera placement and deep-learning user enrollment. This will include motion-triggered video recording and automated event snapshots for household security.
- **Intelligent Baby-Care (Acoustic AI):** Refine the ML model to classify multiple infant states (hunger, sleepiness, playfulness) while implementing noise-cancellation to reduce false triggers from household background sounds.
- **Smart Aquarium Expansion: Automated Correction:** Integrate dosing pumps to automatically adjust pH and nutrient levels based on sensor readings.
    - **Fish Welfare AI:** Use the camera feed to monitor fish movement patterns, using ML to detect early signs of stress or illness.
    - **Automated Cleaning:** Implement a solenoid-based water drainage and replenishment system triggered by high turbidity (TDS) levels.
- **Sustainability & Analytics:** Deploy non-invasive energy monitoring for household appliances to provide monthly power usage analytics and cost-saving recommendations.

11.Conclusion

This proposal presents a comprehensive **Smart Home prototype** that bridges the gap between basic automation and intelligent care by integrating **safety monitoring, access control, automated aquarium management, ML-powered baby care, and real-time CCTV surveillance** into a single ecosystem.

The **dual-processor architecture (ESP32 and Raspberry Pi)** ensures the system remains responsive to real-time events while capable of high-level data processing and AI inference. Successful implementation demonstrates a scalable IoT framework that significantly improves daily convenience, household safety, and peace of mind for both caregivers and pet owners.